



Fact Check: Mitragynine Liver Conversion to Natural 7-OH Is as Dangerous as Synthetic 7-OH

THE CLAIM. Organizations and individuals advocating for blanket kratom bans contend that natural leaf kratom products are as dangerous as products with isolated, concentrated, synthetic alkaloids such as synthetic 7-hydroxymitragynine (7-OH). They argue that twenty percent of the mitragynine in a natural leaf kratom product is converted into natural 7-OH, so by quadrupling the dose of mitragynine you get the same effects as a dose of a concentrated synthetic 7-OH product.

THE TRUTH. The available evidence strongly suggests that while a portion of the mitragynine in natural leaf kratom converts in the body into natural 7-OH, it is nowhere near a 20% conversion rate, and the available literature shows that natural leaf kratom does not pose the same magnitude of risk to consumers as concentrated synthetic 7-OH.

Evidence That Disproves the Claim

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THE CONVERSION RATE IS FAR BELOW 20%. Over 60% of the alkaloid fraction found in natural leaf kratom is mitragynine, while less than 2% is natural 7-OH. Mitragynine is a very weak mu-opioid receptor stimulator¹ while isolated 7-OH has potency like traditional opioids.² While it is true that some mitragynine does convert inside the liver into natural 7-OH, this claim is untrue for many reasons.

Saying that 20% of a mitragynine dose is turned into natural 7-OH because the concentration ratio of natural 7-OH to mitragynine at peak (C_{max}) or over time (AUC) in the blood is roughly 0.2 is very likely to be inaccurate. Under that false equation, 50 mg of mitragynine $\times$ 0.2 = 10 mg of 7-OH. However, in actuality:

- It is impossible to calculate the amount of 7-OH with certainty without the volume of distribution (Vd) being known, and that cannot occur in studies assessing conversions.
- Only animal data provides an estimate of the Vd, which must then be scaled up to humans (which is imprecise).
- If we assume the rat Vd of 2.7 L/kg holds when scaled up to a 70 kg human (189 L), we can estimate the amount of 7-OH created.
- Systemic amount = (AUC)(CL), where $CL = (0.693 \times Vd) / t_{1/2}$.
- Using human single-dose 7-OH pharmacokinetic data from Huestis 2024 for a 53.2 mg mitragynine dose (C_{max} 22.7 ng/mL, AUC_{0-24} 125 ng·h/mL, and $t_{1/2}$ 4.7 h from Table 4) and the scaled-up rat Vd (189 L), the estimated amount is 3.5 mg.
- To back-calculate: 3.5 mg of 7-OH divided by the estimated rat Vd of 189 L gives an estimated C_{max} of 18.5 ng/mL, close to the 22.7 ng/mL in Huestis 2024.
- It is best not to identify a specific amount, though, since confidence around the 189 L Vd is uncertain. As the Vd ranges from 100 L to 250 L, the amount of 7-OH would range from 1.8 to 4.6 mg.
- This means a ~50 mg dose of mitragynine (3.3 grams of generic natural leaf powder) would likely yield a total amount of natural 7-OH created over time between 1.8 and 4.6 mg.

- Since natural 7-OH is created more slowly but cleared more quickly than mitragynine, only a small number of milligrams of 7-OH exists in the body at any one time — which is also different from what would happen if 7-OH were directly administered.

02

ANIMAL STUDIES SHOW MITRAGYNINE DOES NOT BEHAVE LIKE CONCENTRATED SYNTHETIC 7-OH. In pre-clinical studies, purified mitragynine is administered to animals that also metabolize some of it into natural 7-OH, so the outcomes observed already include the contribution of what is converted to natural 7-OH in the body.

- **Respiration:** Three animal studies assessing respiration after mitragynine administration showed no dose-dependent reductions in breathing like the morphine or OxyContin they were studied against. Two also compared concentrated synthetic 7-OH, whose respiratory depression was dose-dependent and approximated morphine's. If the claim were true, the mitragynine converted into natural 7-OH would have produced the same respiratory depression as directly administered concentrated synthetic 7-OH.^{3,4,5}
- **Addiction:** When morphine-addicted animals were switched to mitragynine, they had reduced morphine intake after it was reintroduced (even though some mitragynine converted to natural 7-OH), and mitragynine given alone showed no reinforcing addictive effects. Animals switched to concentrated synthetic 7-OH had increased morphine intake, and concentrated synthetic 7-OH alone showed the same reinforcing addictive effects as traditional opioids.⁶
- **Brain concentration:** At equipotent pain-relieving doses on hotplate testing, the natural 7-OH brain concentration created by mitragynine metabolism was 11-fold lower than after concentrated synthetic 7-OH was administered directly — suggesting natural 7-OH is one of, but not the main, mechanism of pain relief for the natural leaf.⁷
- **Dose equivalence:** To reach the same central-nervous-system natural 7-OH concentrations as giving synthetic 7-OH alone, researchers had to give 200 times the mitragynine dose (140 mg/kg) to match a concentrated synthetic 7-OH dose (0.7 mg/kg), owing to low conversion rates and differences in volume of distribution.⁸

03

HUMAN STUDIES SHOW NATURAL LEAF KRATOM LACKS TRADITIONAL-OPIOID EFFECTS. Human studies clearly suggest the magnitude of the opioid effects of natural leaf kratom is not like that of traditional opioids.

- In an FDA-conducted clinical trial, natural leaf kratom — even at doses (12 g) four times higher than usually taken by consumers (1–3 g) — was associated with only about half the pupil constriction expected from traditional opioids. Pinpoint pupils (severe constriction) are a marker of the potency of opioid-receptor stimulation.⁹
- In a descriptive study, one of the most common reasons cited for use was that consumers believed natural leaf kratom was much less dangerous than other substances. Unlike illicit opioids, which compromise people's ability to conduct activities of daily living, 70–80% of consumers identified natural leaf kratom as helping with these activities; in another study, 54% said it actively helped them complete daily activities and 84% felt it was at least not a hindrance.¹⁰

04

PHARMACOLOGY EXPLAINS THE DIFFERENTIAL EFFECTS. There are many potential reasons for the differential effects between natural leaf kratom products and concentrated synthetic 7-OH products:

- **Entourage effect:** Attributing all of natural leaf kratom's effects to mitragynine and natural 7-OH is an oversimplification — there are a dozen biologically active alkaloids working on serotonin and alpha-adrenergic systems, anti-inflammatory mechanisms, and opioid-receptor blocking (among others), complementing and working against one another. Isolated concentrated synthetic 7-OH contains none of these and has no such entourage effect.¹¹
- **Multiple pain mechanisms:** This is why, at equipotent pain-relieving doses, the brain concentration of 7-OH is much lower with mitragynine use than when isolated concentrated synthetic 7-OH is given — natural leaf kratom relieves pain through multiple mechanisms.
- **Onset and euphoria:** Opioids that reach high peak concentrations and clear quickly produce more euphoria and faster tolerance and dependence. Taking concentrated synthetic 7-OH directly in an isolated product likely produces a faster onset and higher peak concentrations — which can induce euphoria — and its short time in the body means effects diminish quickly, enhancing the desire to redose.
- **Kinetics:** In animals, peak concentration occurs ~20 minutes after concentrated synthetic 7-OH is given directly.¹² In humans given mitragynine, the peak occurs ~1 hour afterward, but natural 7-OH does not peak until ~1.7 hours later.¹³ Combined with the very short ~30-minute half-life when concentrated synthetic 7-OH is given directly to animals, versus a 3–6 hour half-life in humans when natural 7-OH is formed from mitragynine, the peak natural 7-OH effect is delayed, lower, and diminishes more slowly — because mitragynine keeps turning into natural 7-OH and replenishing the supply.
- **Competitive antagonism:** Mitragynine is far less potent as an opioid than 7-OH, and when both are present, mitragynine may interfere with natural 7-OH's ability to stimulate opioid receptors — shown by its ability to blunt morphine's effect in animals.¹ Natural leaf kratom has abundant mitragynine available to temper the effects of the natural 7-OH being created, again reducing the chance of euphoria.
- **Enzyme saturation:** With standard natural leaf or mitragynine dosing, the naturally produced 7-OH concentration is about 20% of the mitragynine concentration.¹³ However, that ratio goes down as the mitragynine dose goes up — the metabolic enzymes become saturated, so more mitragynine is eliminated without converting to natural 7-OH¹⁴ — and it also goes down after multiple doses versus single doses.¹⁵

What National and International Bodies Say

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REGULATORS DISTINGUISH THE TWO PRODUCTS. National and international regulatory and health bodies have articulated different determinations for isolated concentrated synthetic 7-OH products and natural leaf kratom products:¹⁶

- HHS and the FDA have recommended to the DEA that concentrated synthetic 7-OH products be made Schedule I (banned) but that natural leaf products should not, finding a clear difference in potential societal risk between the two.¹⁷
- The American Medical Association believes a clear difference exists in the risk profiles of concentrated synthetic 7-OH products and natural leaf kratom products: it supports mak-

ing concentrated synthetic 7-OH Schedule I but is instead calling for additional clinical trials of natural leaf kratom and mitragynine products.¹⁸

- The FDA released its first-ever clinical trial of natural leaf kratom without noting concerning findings, and NIDA and the FDA are funding additional studies of natural leaf kratom and its main alkaloid mitragynine. Studies of this type are not planned at the federal level for concentrated synthetic 7-OH products.^{19,20}
- The DEA and WHO assessed natural leaf kratom years ago and decided a ban was not clearly in the public interest, and have expressed no interest in revisiting those determinations since.¹³

Conclusion

The KCAC has shown that this claim is based on a nugget of truth but disregards multiple additional factors that render it untrue. Whether these claims are made from ignorance of the relevant pharmacokinetics or pharmacology, or stated intentionally to manipulate the public, legislators, and regulators, is unknown. However, the KCAC strongly believes that untrue, sensationalist, sweeping claims make it harder to engage in meaningful discussions with important stakeholders from which reasonable solutions can be found. Please see our fact-checking documents, position statements,²¹ and an 8-Factor analysis¹⁶ for more information on this complex landscape.

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